
From Trash to Tensors
Local AI "On A Budget" 🤡

$ whoami



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→ **Security Professional:** 12+ years in software security. OSCP, CISSP, CCSKv4, ISO 27001/42001

→ **AI/ML Enthusiast:** MSc in Data/ML for fun, hobby projects in machine listening, agentic SWE and security

→ **Local AI:** Started with my gaming rig a few years ago and now I'm running my own cluster. Have run inference from phones and tablets to data center GPU clusters.

Always available for chats about security and synthesizers! 🎛️

Reality Check



LOCAL AI IS EXTREMELY **EXPENSIVE** DUE TO COMPONENT SHORTAGES ⚠

→ **Cheapest way:**
Travel back in time - most of this hardware was way cheaper

→ **Use AI inference:**
API inference will **most likely** be cheaper than local for hobby

→ **Pick your poison:**
- Modern hardware is *very* expensive
- Old hardware is unreliable, loud, wasteful and expensive
- Cheap hardware is usually slow, limited or unsupported

→ **Memory is key, so is patience:**
As long as you have enough RAM you can run LLMs, but expect the speed of old dial-up modems.

Local AI Potential Bottlenecks and Problems (Learned the Hard Way)

→ **GPU:**
Available VRAM, Speed, Power, AI features, Software support

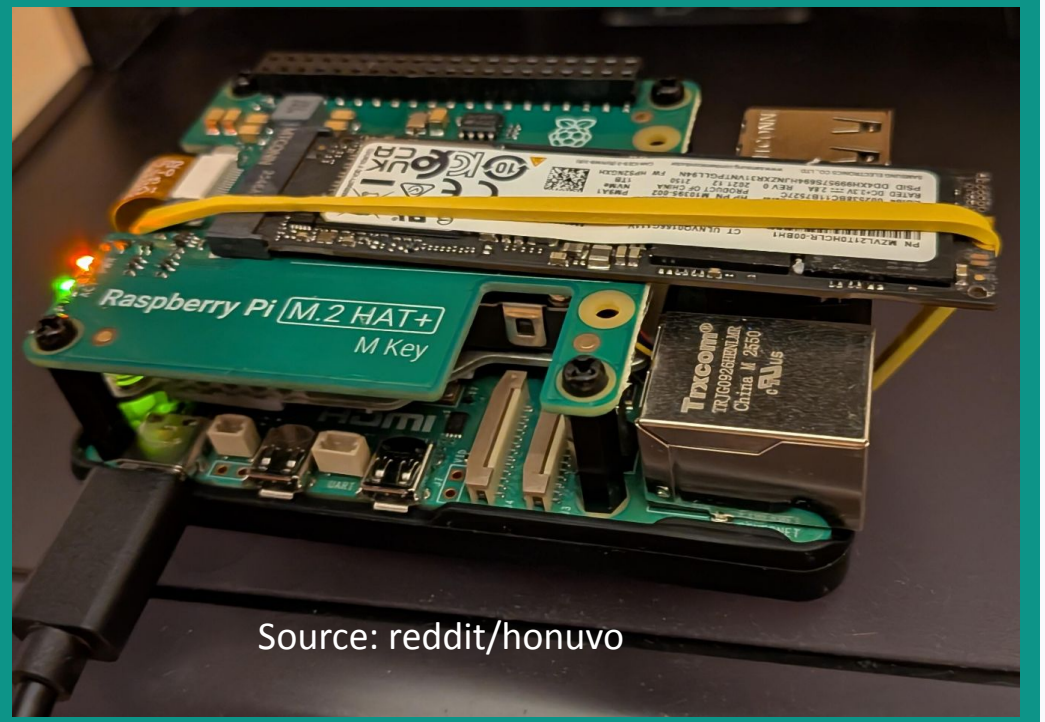
→ **Memory:**
Available RAM, Speed, Error Correction

→ **Motherboard:**
PCIe lanes (GPU), Memory Channels (RAM)

→ **PSU, Electrical Wiring, Physical Space, Noise, Electricity**
Multiple GPUs may require beefy PSUs, electrical wiring and approval from partner, family and other stakeholders

The Local AI Scavenger Lifecycle





Single Board Card Computer

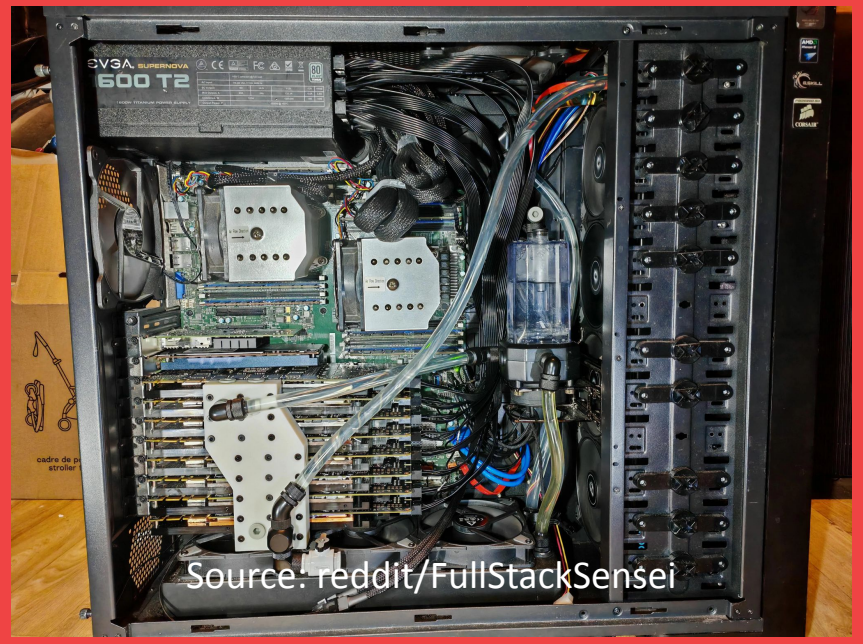
~200,000 Ft (€550)

Model: Qwen2.5-1.5B-Instruct
Speed: ~20-35 tokens/sec

Hardware Specs & Price Breakdown

Component	Exact Hardware Model	Price
System / CPU	Raspberry Pi 5 (8GB RAM) Raspberry Pi 5 (16GB RAM)	**70,000 Ft/** **120,000 Ft**
NPU Accelerator	Raspberry Pi AI Hat 2	**100,000 Ft**
Storage	512GB NVMe M.2 2242 SSD	**20,000 Ft**
Power & Case	Case + USB-C PD Supply	**10,000 Ft**

→ ***Only*** useful if your goal is edge AI
→ Unpopular model size (1.5B)
→ No major model release since 2025
→ Expect all models to be slow and limited



Source: reddit/FullStackSensei

Data Center Dumpster Dive 🗑️

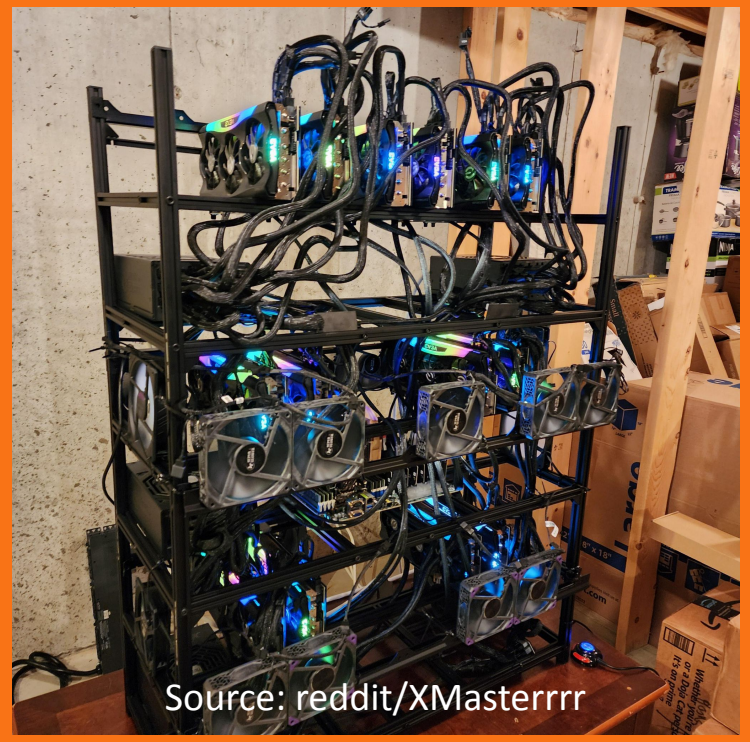
~620,000 Ft (€1,700)

Model: Qwen3.8-Flash-Next-IQ2-XS
Speed: ~37 tokens/sec

Hardware Specs & Price Breakdown

Component	Exact Hardware Model	Hardverapró Est.
GPU / VRAM	2x Nvidia Tesla P40 24GB GDDR5 (48GB total)	**300,000 Ft**
Motherboard	X570 or Z690 (with x8/x8 PCIe splitting)	**65,000 Ft**
CPU & RAM	AMD/12th Gen Intel + 64GB DDR4-3200	**200,000 Ft**
Power & Cooling	850W Gold ATX PSU + Case + 3D Fan Shrouds	**55,000 Ft**

→ Decent speed and VRAM size for mid-tier models
→ 10 year old data center hardware
→ Requires custom 3D printed shrouds and watercooling
→ Lack of support for modern AI features (eg. FP quants)



Ex-Crypto Frankenrig 📉

~1,500,000 Ft (€4,100)

Model: Qwen3-Coder-Next (80B EXL2)

Speed: ~25 tokens/sec

Hardware Specs & Price Breakdown

Component	Exact Hardware Model	Hardverapró Est.
GPU / VRAM	Up to 6 RTX 3/4/5060+ class GPUs (72-96 GB VRAM)	**850,000-1,200,00 Ft**
CPU	Intel Celeron G4900 / Core i3-8100 (LGA1151)	**15,000 Ft**
Motherboard	Onda B250 / BTC-D1800 6-slot Mining Board	**45,000 Ft**
Memory	32GB (2×16GB) DDR4-2666MHz Non-ECC	**25,000 Ft**
Power & Frame	2× 1200W ATX PSUs + Open-air Mining Frame + PCIe x1 Risers	**215,000 Ft**

→ Repurposed BTC mining rigs have plenty of GPU slots
→ PCI-E 3.0 1x risers bottleneck performance
→ Slow TTFT, but decent-ish token generation
→ Good for Dense models, bad for MoE



Source: reddit/nilesism

Consumer Modern GPU 🎮

~1,040,000 Ft (€2,750)

Model: ISTA-DASLab/Qwen3.8-Flash-Next

Speed: 72 tokens / second

Hardware Specs & Price Breakdown

Component	Exact Hardware Model	Hardverapró Est.
GPU / VRAM	2x NVIDIA RTX 5060 Ti 16GB (32GB Total, GDDR7)	**480,000 Ft**
Motherboard	MSI MAG X670E Tomahawk WIFI (PCI-E x8/x8 split)	**60,000 Ft**
CPU	AMD Ryzen 9 7900	**160,000 Ft**
Memory	64GB (2×32GB) DDR5-6000MHz	**240,000 Ft**
PSU & Case	850W ATX 3.0 PSU + Mid-Tower Chassis	**100,000 Ft**

→ Hobbyist favorite
→ Logical upgrade from a gaming PC
→ High-end motherboard + dual GPU + extra RAM
→ Can upgrade GPU or RAM for better AI performance
→ **Can easily be sold as well**



Workstation 🛠️

~3,750,000 Ft (€ TBD)

Model: Qwen3-Coder-30B-A3B-Instruct

Speed: ~85 tokens/sec

Hardware Specs & Price Breakdown

Component	Exact Hardware Model	Hardverapró Est.
GPU / VRAM	NVIDIA RTX 6000 Ada Generation 48GB	**3,000,000 Ft**
CPU	AMD Ryzen 9 7950X	**190,000 Ft**
Motherboard	MSI MAG B650 Tomahawk WiFi ATX	**85,000 Ft**
Memory	64GB (2×32GB) DDR5-6000MHz	**240,000 Ft**
PSU & Case	1000W Seasonic PCIe 5.0 ATX 3.0 + High-airflow Tower Case	**220,000 Ft**

→ Single fast GPU + lot of VRAM choice
→ No driver or setup complications
→ Dense ~30B model + roomy KV cache
→ Perfect for local training, fine-tune of smaller models
→ Needs more RAM and better motherboard for big models



Unified RAM Machines 🤝

~250k Ft – ~5+ Million Ft (€1k–€5.3k+)

Model: Whatever you want

Speed: 120 token/sec →15 token/sec

Hardware Specs & Price Breakdown

Vendor	Exact Hardware Model	RRP
AMD	Strix Halo / AI Max 395+ 128GB	**~2,000,000 Ft**
Apple	Mac Mini/Studio 16GB-256GB	**~250,000 Ft - ~5,000,000 Ft**
Nvidia	DGX Spark and OEM models 128GB	**~2,750,000 Ft**

→ Small machines with decent local AI performance
→ Can be clustered ***or*** expanded via eGPU
→ Fits into living room, under desk or into 10"/19" racks
→ 4+ clustered DGX Sparks can host larger LLMs
→ Low power consumption
→ Performance is "acceptable" for hobby use
→ "Upgrade" means buying another one *or more*


Source: Reddit/Zigong_actias

CPU+RAM Only 🤖

~10,600,000 Ft (€29,000)

Model: GLM-5.3_Q4_K_M

Speed: < 10 tokens / second

Hardware Specs & Price Breakdown

Component	Exact Hardware Model	Hardverapró Est.
Memory (Main Cost)	1.5TB (24× 64GB) DDR5-4800MHz ECC	**4,800,000 Ft**
CPU Sockets	2x AMD EPYC 9654	**3,200,000 Ft**
Server Platform	Supermicro / Gigabyte Dual-Socket SP5 2U Barebone	**2,600,000 Ft**
GPU / Accelerators	LOOK MOM, NO GPU 😱	**0 Ft**

→ Can run frontier models "on a budget"
→ Large context and KV cache
→ Loud, high power consumption
→ Slow generation and pre-fill
→ Good for batch and async when latency doesn't matter

tl;dr

→ No truly affordable way to do local AI
→ API inference or renting is a valid alternative
→ Old rigs are cheap because of limitations
→ Get as much RAM as possible
→ Best hardware for local AI is the one you already have

Thank You



You can find the slides here.